how to make a project into a (pleasant) respected instrument
best sounding music to the ear is a logarithmic one

$y = \log_{(c)}(x)$

TITLE

Air Theremin

TOPIC

Build Guide: from a blinking LED to playing music with your hands

DEPARTMENT

IEEE SBUB · Project Resurrection

FULL NAME

Youcef HARMOUNI

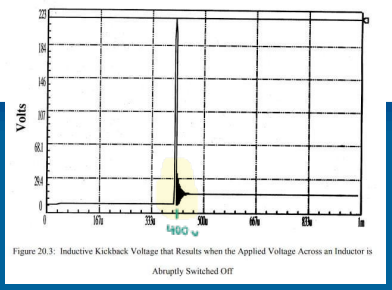


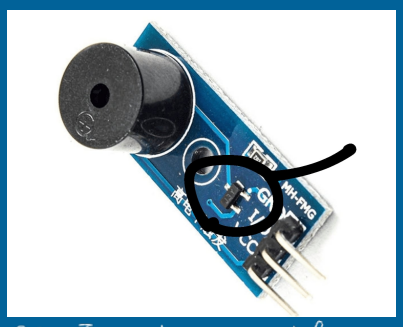
Figure 20.3: Inductive Kickback Voltage that Results when the Applied Voltage Across an Inductor is Abruptly Switched Off

but cuz of Resistance of the wire ...
it reduces to just a high value
in range of few 100s to 1000s Volt
(still enough to burn your Arduino)

need to protect the circuit : That's why
we need A MOTOR DRIVER to drive simple DC motor

What's a motor drive?
is an electronic circuit that lets a low-power controller, like an Arduino control a motor that needs more current

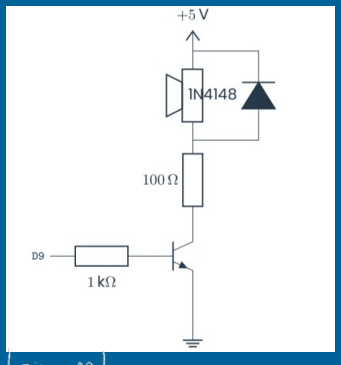
Arduino sends control signals → Motor driver controls motor power & helps protect the Arduino → DC motor Receives the power it needs



e.g.: In a buzzer module,

when it comes to BOOM
The diode take the whole charge
(simple diode can handle 1000 v)

we have this diode protective circuit

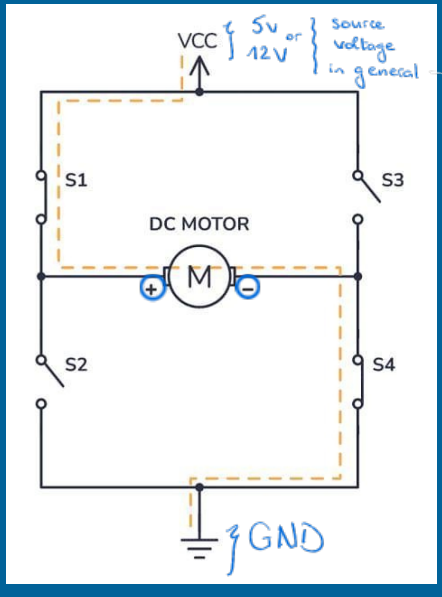


pic p.27

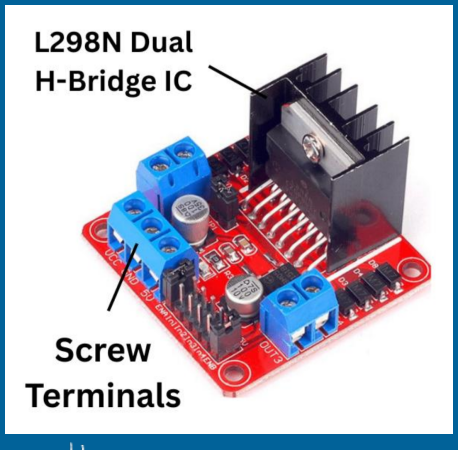
Extra note:
* A simple Transistor has a bunch of diodes as the first
* Second transistor has something called "cut off", this Cutoff make it the circuit sort of Open

100 or 1000 doesn't matter it's Open

like Arduino one



Voltage matters but so is current



H-bridge

☆ If we close switch 1 & 4 the motor gets voltage from its ⊕ → makes the motor spins forwards ↻

Because the ⊖ side of the motor needs a path to the GND = ground, negative of the battery

☆ For backwards, we turn on switch 2 & 3, so that the motor goes in from the ⊖ of the motor

Reason of using it: → inductive Kickback
→ Arduino gives current in few mA where the motor needs it in the 100s mA

So we need a power circuit that deliver it properly
(it has a circuit inside that "t3sr" the battery to deliver more current)

Assignement

① Make a simple blinking LED
- 1s off
- 1s on

Extra: Try now making two LEDs blinking
- one blue
- one red

in which
- 1st s: blue on, red off
- 2nd s: blue off, red on

② Make a LDR circuit

Light Dependant Resistor

⇒ It is specified in the document

* Try it
* Try moving the light
* Make it turn on LED after a threshold

limit or level that something must reach before an action happens

Extra: Try making a LED pulsing with LDR
Meaning it kinda becomes dim when it is bright

less bright

Hint: The LDR reads from 0 - 1023
But your analog write works from 0-255
→ so you need a way to convert it
⇒ you can try two methods:
- dividing by 4
- use this built in function called map()

Read about it here:

Use the website Tinkercad =D!

docs.arduino.cc